ACOSTA RESIDENCE

New single-family residence, Eaton Fire rebuild · 220 Laurel Dr, Altadena CA 91001

3D massing, from the street (for reference)



SCOPE OF WORK
New one-story, all-electric single-family residence (4 bedrooms, 3 baths) with an attached open carport, replacing the house destroyed in the Eaton Fire. Demolition of the remaining retaining wall and chimney, grading and drainage per civil, new driveway, utilities (water, sewer, electric; no gas). Fire sprinklers throughout.

PROJECT TEAM
Owner: Richard Acosta
Plans prepared by: _____
Structural engineer: _____
Civil engineer: _____
Energy consultant: _____
Fire sprinkler designer: _____

GENERAL NOTES
1. Work shall comply with the codes listed and the conditions of approval.
2. Do not scale drawings; dimensions govern. Verify on site.
3. Structural, civil and energy documents govern their own scope.
4. Special inspections and structural observation per the structural engineer.
5. Smoke and CO alarms per A-2; fire sprinklers per the deferred submittal.

SHEET INDEX

SHEET	TITLE
A-0	Cover sheet, code analysis
A-1	Site plan, project data
A-2	Floor plan, door and window schedules
A-3	Roof plan
A-4	Exterior elevations
A-5	Building sections
A-6	Fire-zone details (CRC R337)
A-7	Attic and structural loads plan
S-	Structural (by the structural engineer)
C-	Grading and drainage (by the civil engineer)
T-24	Energy compliance (CF1R) and solar PV (by the energy consultant)
CALGreen	Green building checklist

DEFERRED SUBMITTALS

#	ITEM
1	Fire sprinkler plans and calculations (NFPA 13D), before the final foundation inspection
2	Roof truss drawings and calculations (truss manufacturer), reviewed by the engineer of record
3	Solar PV plans, if not in the energy package

CODE ANALYSIS

ITEM	PROJECT
Codes	2022 California Residential, Building, Electrical, Mechanical, Plumbing, Energy, Fire and Green Building Standards Codes, as allowed for fire rebuilds by Executive Order N-29-25; LA County Title 26 amendments
Occupancy	R-3 single-family dwelling; U open carport (attached)
Construction	Type V-B; automatic fire sprinklers throughout (NFPA 13D)
Stories / height	1 story; ~17'-6" above finished floor (30' limit, Altadena CSD)
Areas	Dwelling 2,238 sq ft · porch 37 · carport 396 sq ft
Lot / coverage	7,319 sq ft · 2,672 sq ft (36.5%)
Zoning	R-1-7500 · Altadena CSD · Disaster Recovery Permit (non-like-for-like)
Fire zone	VHFHSZ: wildland-urban interface construction, CRC R337 (A-6)
Energy	All-electric; Title 24 Part 6 compliance by the energy consultant
Green building	CALGreen mandatory measures (checklist)

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220 Laurel Dr
Altadena, CA 91001
APN 5832-003-027

Zone R-1 · Altadena CSD
Very High Fire Hazard
Severity Zone

Owner: Richard Acosta

Prepared by: _____

Structural: by engineer
(separate sheets)

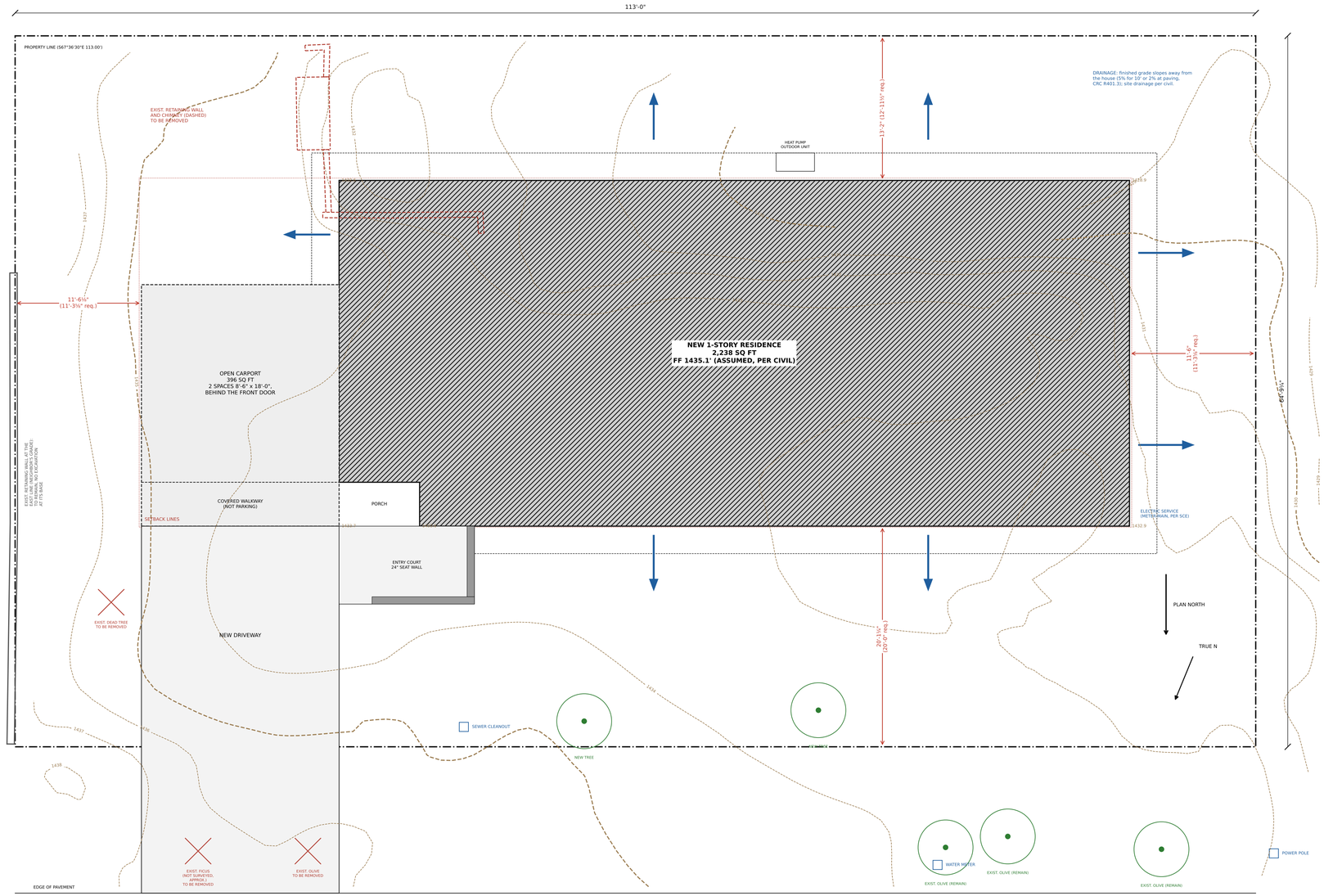
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**COVER SHEET,
CODE ANALYSIS**

A-0



LAUREL DRIVE (50' R/W)

SITE PLAN 3/16" = 1'-0"

PROJECT DATA

Scope	New 1-story single-family residence with attached open carport; Eaton Fire rebuild (non-like-for-like). Demolish remaining retaining wall and chimney.
Address / APN	220 Laurel Dr, Altadena CA 91001 APN 5832-003-027
Zoning	R-1-7500 · Altadena CSD
Fire zone	Very High Fire Hazard Severity Zone
Occupancy	R-3 (dwelling), U (carport)
Construction type	V-B, fully sprinklered (NFPA 13D)
Codes	2022 California Residential, Building, Electrical, Mechanical, Plumbing, Energy and Green Building Codes (fire rebuild, EO N-29-25), as amended by LA County Title 26
Lot	113.00' x 64.77' = 7,319 sq ft
Floor area	2,238 sq ft (to outside of walls)
Porch / carport	37 / 396 sq ft
Lot coverage	2,672 sq ft (36.5%); limit 2,830
Height	1 story, −17'-6" above finished floor
Bedrooms / baths	4 / 3
Parking	2 covered (carport)

SITE NOTES

1. Lot lines, existing trees, utilities, spot elevations and 1' contours (dashed) from the owner's topographic survey.
2. Finished floor elevation, grading and drainage per civil (separate sheets).
3. Eaves (dashed) 2'-6" all round (LACC 22.110.090).
4. West parkway olives protected during construction (root zones); the dead lot tree and the east parkway olive and ficus removed (County permit).
5. Ember-resistant zone under the eaves and 5' beyond them; planting per the fuel modification plan (FM-1).
6. Water, sewer and SCE service per the utilities; no gas service (all-electric).
7. Demolition: remove the remaining retaining wall and chimney from the burned house (dashed red); asbestos survey first if required (SCAQMD Rule 1403); fill and grade per civil.
8. The retaining wall along the east line holds the neighbor's higher grade: protect it in place.

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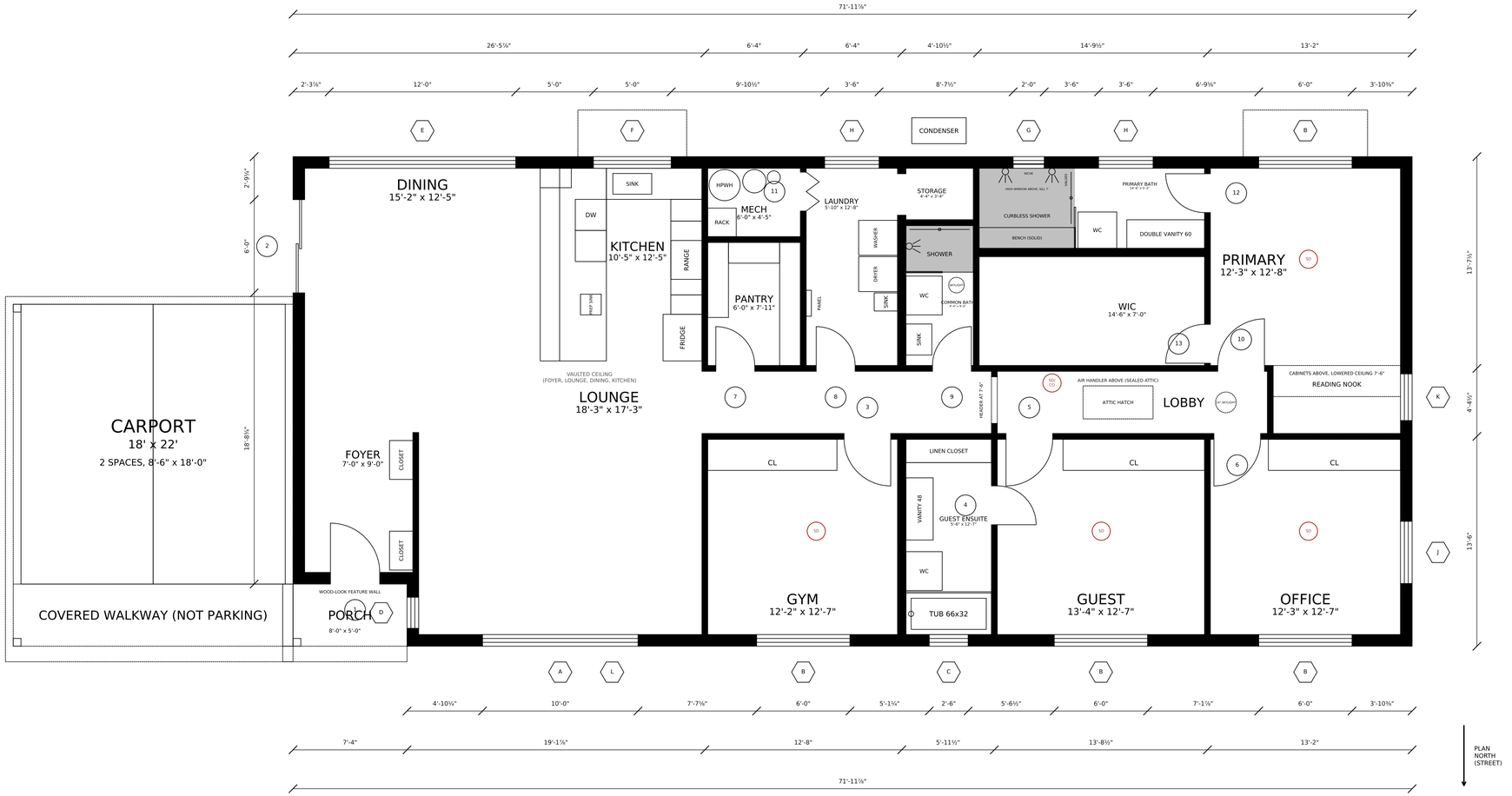
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**SITE PLAN
& PROJECT DATA**

A-1



FLOOR PLAN 1/4" = 1'-0"

GENERAL NOTES

1. Dimensions are to wall centrelines; exterior walls to the outside face of framing + finish (9" walls).
2. Walls: exterior 2x6 with 1" rigid insulation and stucco; interior 2x4; plumbing walls 2x6.
3. Ceilings 9'-0" unless noted. The foyer, lounge, dining and kitchen are vaulted under the roof (see sections).
4. Doors 8'-0" high unless noted; interior doors solid core. Exterior doors and glazing per CRC R337 (VHFHSZ).
5. All exterior glazing dual-pane with at least one tempered pane (CRC R337.8.2.1). Fully tempered where noted (CRC R308.4).
6. Bedroom windows meet emergency escape: 5.7 sq ft net clear, 20" min width, 24" min height, sill 44" max (CRC R310).
7. Smoke alarms: hard-wired, interconnected, battery backup, at least 3' from bathroom doors (CRC R314). Combination smoke / CO alarms outside the sleeping areas.
8. All-electric: heat pump HVAC (air handler in the sealed attic; outdoor unit at the rear wall), heat pump water heater (HPWH) in the mechanical closet. Whole-house ventilation by a continuous exhaust fan.
9. Fire sprinklers throughout (NFPA 13D); plans by the sprinkler designer (deferred submittal).
10. Attic access in the hall: 22" x 54" insulated, gasketed access panel (passes the air handler), 30" min headroom above (CRC R807.1).
11. Furniture not shown; closets, cabinets and fixtures shown for reference.

DOOR SCHEDULE

TAG	LOCATION	SIZE (W x H)	TYPE	NOTES
1	PORCH / FOYER (exterior)	3'-0" x 8'-0"	Hinged	Solid core 1-3/8"+ or noncombustible (CRC R337.8.3); weatherstripped; glass lite, if any, dual-pane tempered
2	DINING (exterior)	6'-0" x 8'-0"	Sliding glass door	Dual-pane tempered glazing (CRC R308.4.1, R337.8.2); weatherstripped
3	GYM / LOBBY	3'-0" x 8'-0"	Hinged	
4	GUEST ENSUITE / GUEST	2'-6" x 8'-0"	Hinged	
5	GUEST / LOBBY	3'-0" x 8'-0"	Hinged	
6	OFFICE / LOBBY	3'-0" x 8'-0"	Hinged	
7	LOBBY / PANTRY	2'-6" x 8'-0"	Hinged	
8	LOBBY / LAUNDRY	2'-6" x 8'-0"	Hinged	
9	LOBBY / COMMON BATH	2'-6" x 8'-0"	Hinged	
10	LOBBY / PRIMARY	3'-0" x 8'-0"	Hinged	
11	MECH / LAUNDRY	2'-6" x 8'-0"	Bifold, louvered	louvered (water heater air)
12	PRIMARY BATH / PRIMARY	2'-6" x 8'-0"	Hinged	
13	WIC / PRIMARY	2'-6" x 8'-0"	Hinged	

WINDOW SCHEDULE

TAG	QTY	LOCATION	SIZE (W x H)	SILL	OPERATION	NOTES
A	1	Lounge	10'-0" x 5'-6"	2'-6"	Fixed center, slider each end	
B	4	Gym, Guest, Office, Primary	6'-0" x 5'-6"	2'-6"	Horizontal slider	Emergency escape (note 6)
C	1	Guest Ensuite	2'-6" x 1'-6"	6'-6"	Awning	Obscure glass
D	1	Foyer	2'-0" x 5'-6"	2'-6"	Fixed, reeded glass	Tempered (CRC R308.4.2)
E	1	Dining	12'-0" x 6'-6"	1'-6"	Fixed, two units mulled	
F	1	Kitchen	5'-0" x 4'-6"	3'-6"	Horizontal slider	
G	1	Primary Bath	2'-0" x 1'-0"	7'-0"	Awning	
H	2	Laundry, Primary Bath	3'-6" x 2'-0"	6'-0"	Awning	
J	1	Office	4'-0" x 2'-0"	6'-0"	Awning	
K	1	Reading Nook	3'-0" x 3'-6"	3'-6"	Awning	
L	1	Lounge	18'-0" x 3'-0"	12'-1½"	Fixed, 3 units mulled	Clerestory

LEGEND

- 1 — Door tag (see door schedule)
- A — Window tag (see window schedule)
- SD — Smoke alarm (SD), smoke / CO alarm (SD/CO)
- --------- Overhead: canopy, attic hatch, skylight

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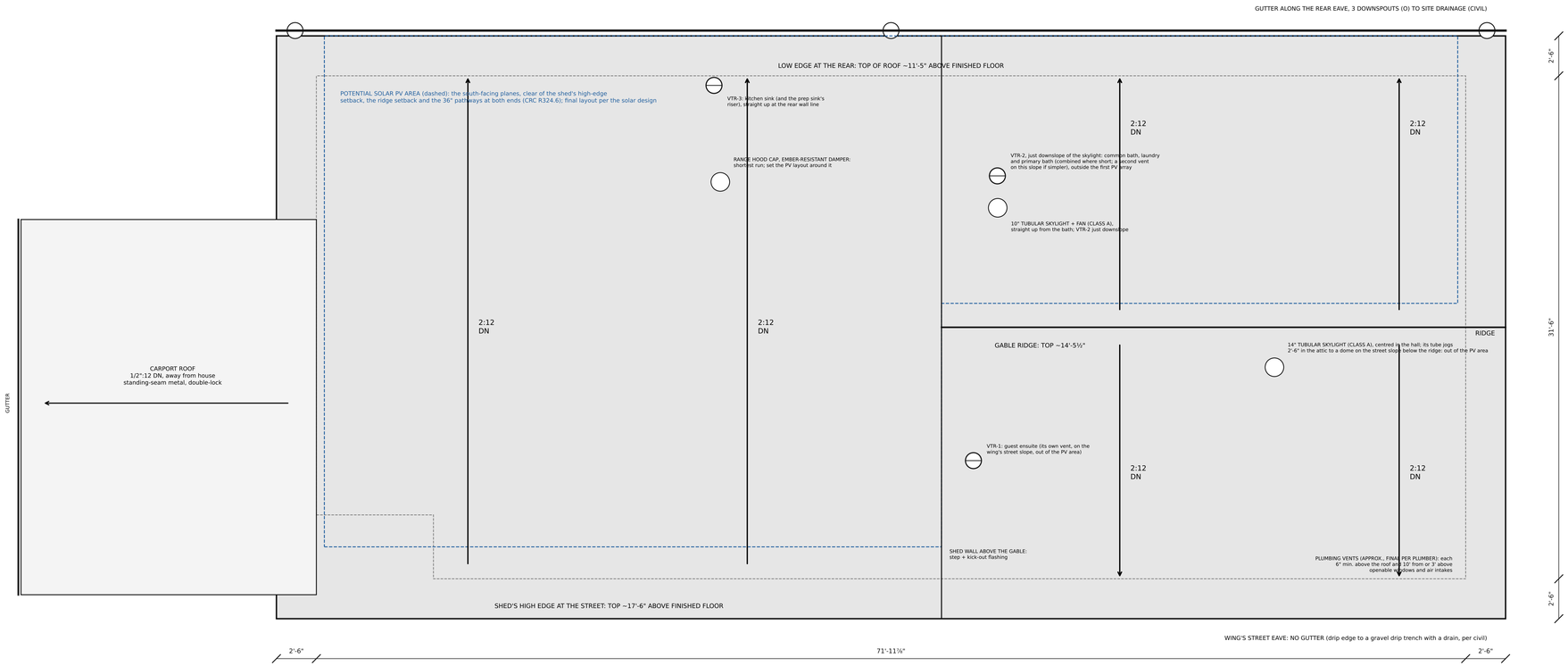
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FLOOR PLAN, DOOR & WINDOW SCHEDULES

A-2



STREET (PLAN NORTH)

ROOF PLAN 1/4" = 1'-0"

ROOF NOTES

1. Main roof at 2:12: a shed from the porch into the bedroom wing, high at the street, and a gable over the rest of the wing. Class A asphalt shingles (fiberglass), dark, wind-rated (ASTM D7158 Class H / D3161 Class F), over two layers of underlayment as required at 2:12 (CRC R905.2); self-adhered underlayment at the eaves and along the shed / gable junction.
2. Sealed (unvented) attic, CRC R806.5: insulation at the roof deck; NO attic or eave vents.
3. Eaves 2'-6" all round, enclosed; dark metal fascia and drip edge. See A-6 for fire-zone details (CRC R337).
4. Rear gutter with noncombustible gutter guards (CRC R337.5.4); downspouts to the site drainage system per civil. No gutter on the wing's street eave: drip edge over a gravel drip trench.
5. Roof framing, sheathing and connections per structural.
6. Roof penetrations flashed per the shingle maker, metal flashings, all outside the first PV array. Plumbing vents (VTR): the guest ensuite's on the wing's street slope; the kitchen's at the rear wall line; the baths and laundry on the wing's rear slope. The hall skylight's dome is on the street slope. Exhaust caps metal with backdraft dampers; bath fans exit through walls where possible. No dryer vent.
7. Carport: attached open carport, its own roof at 1/2":12 in double-lock standing seam (Class A assembly, uplift-rated), gutter on its east edge.
8. Solar: first array (about 10 kW) on the shed, layout by the solar provider. Its mounting (rails over flashed attachments into the rafters) is installed with the roofing; panels later. Its conduit enters through the roof into the attic and runs inside to the panel: none on the roof surface or the exterior walls.
9. Wind: severe downslope winds from the north / northeast strike the shed's high street edge. Wind-rated shingles with the maker's high-wind nailing, sealed starter strips at eaves and rakes; eave and carport tie-downs per structural.

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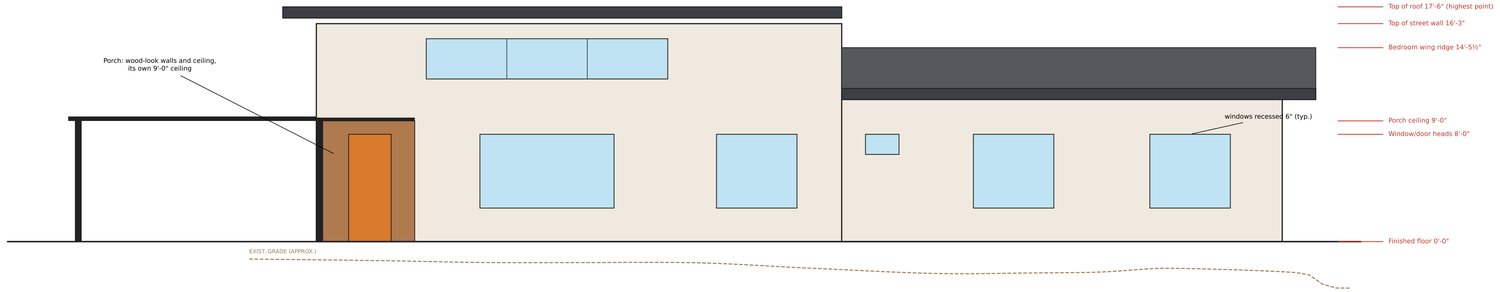
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ROOF PLAN

A-3

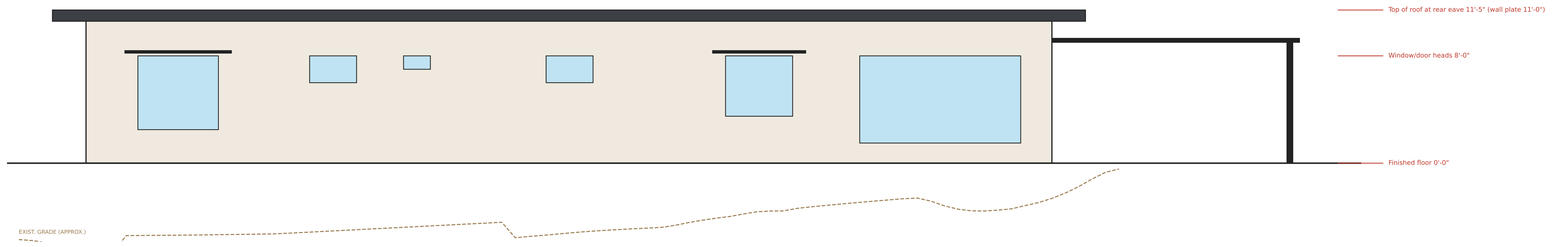
NORTH (STREET) ELEVATION



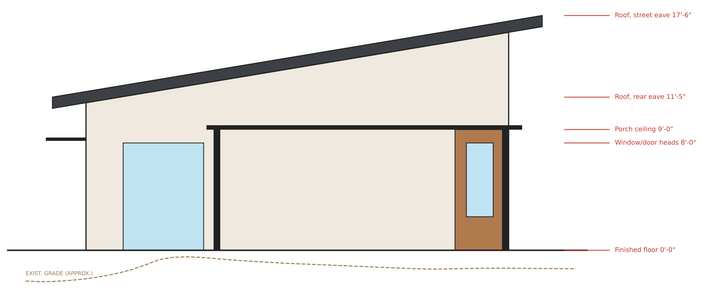
EXTERIOR MATERIALS
1. Walls: 3-coat stucco, light color (noncombustible).
2. Roof: asphalt shingles, dark, Class A (A-3); carport roof standing-seam metal.
3. Fascia, drip edge and canopies: dark metal.
4. Porch walls and ceiling: wood-look fiber cement or porcelain (noncombustible / ignition-resistant).
5. Windows: dual-pane, one pane tempered (CRC R337.8); punched street windows recessed 6" (clerestory band flush). See A-2 schedule.
6. Eave soffits enclosed, no vents (A-6).
7. Existing grade (dashed) is approximate, from the survey with an assumed finished floor; finished grades and floor elevation per civil.

Heights are above finished floor (0'-0"). Assumed finished floor: 1435.1' (6" above the highest existing grade at the walls). Scale: 3/16" = 1'-0"

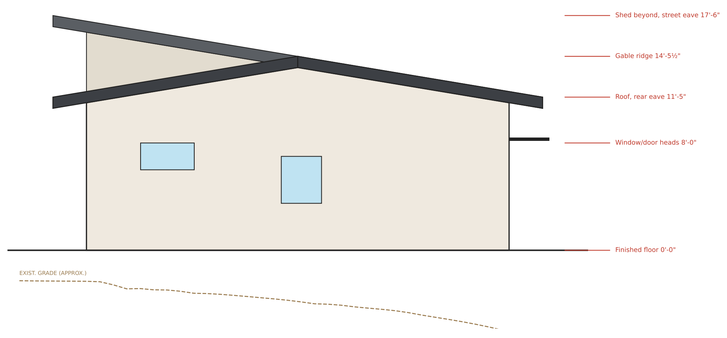
SOUTH (REAR) ELEVATION



EAST (CARPORT) ELEVATION



WEST ELEVATION



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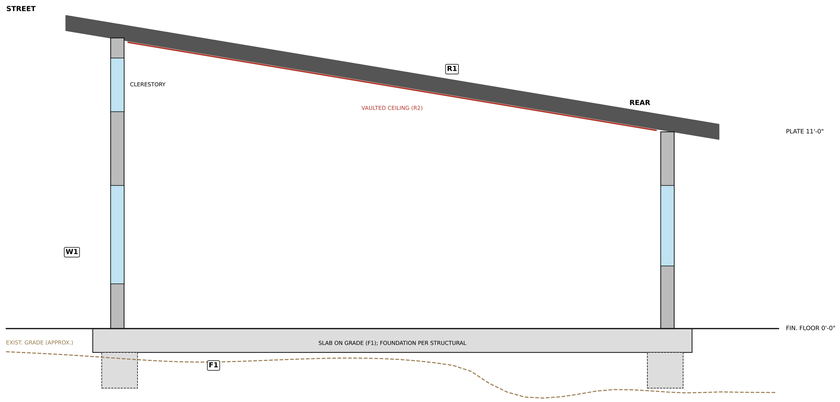
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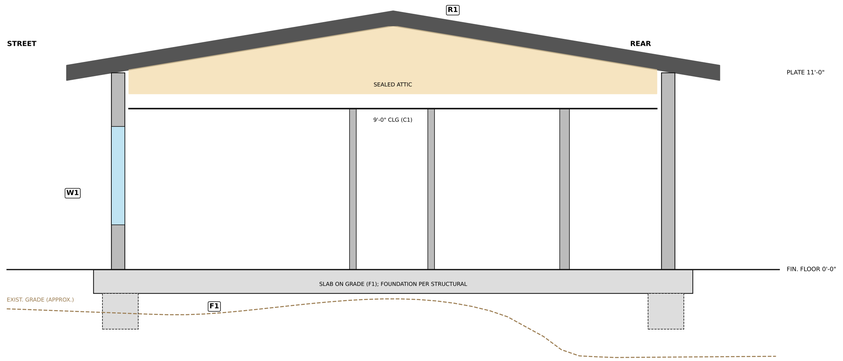
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EXTERIOR ELEVATIONS

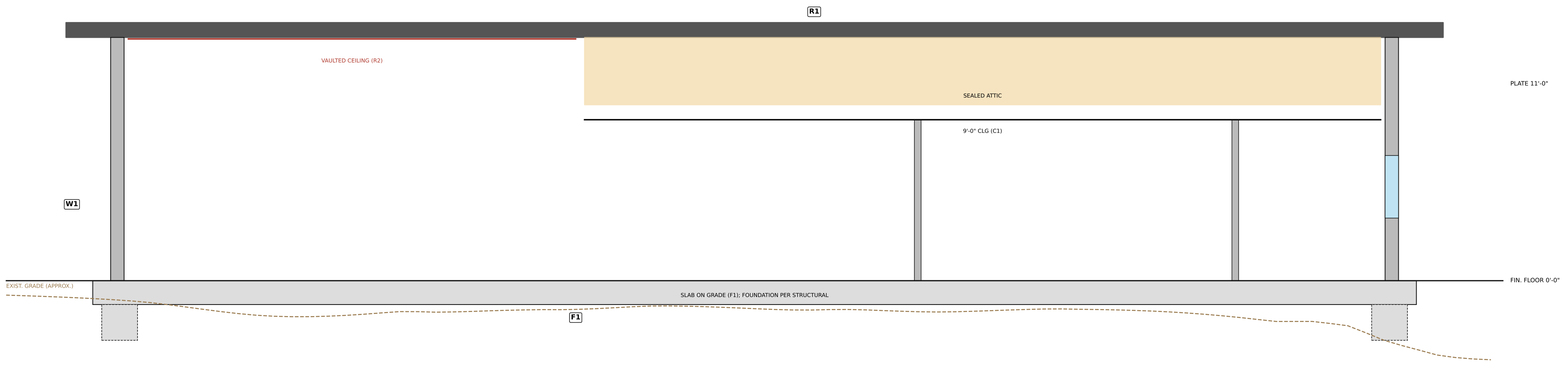
A-4



SECTION A: THROUGH THE LOUNGE AND KITCHEN (VAULTED) 1/4" = 1'-0"



SECTION B: THROUGH THE GUEST ROOM, LOBBY AND WALK-IN CLOSET (GABLE) 1/4" = 1'-0"



SECTION C: ALONG THE LOBBY, LOOKING SOUTH 1/4" = 1'-0"

ASSEMBLIES (insulation values per the Title 24 report)
R1 ROOF: Class A asphalt shingles on two layers of underlayment (2:12), roof sheathing, framing per structural; spray foam at the underside of the deck (unvented, CRC R806.5); ignition barrier / thermal barrier where the attic is used.
R2 VAULT: as R1, gypsum board on the framing (no attic).
C1 CEILING (9'-0"): 5/8" gypsum board; sealed attic above.
W1 EXTERIOR WALL: 7/8" 3-coat stucco, weather barrier, 1" rigid insulation, sheathing, 2x6 studs with cavity insulation, 1/2" gypsum board.
W2 INTERIOR WALL: 2x4 (plumbing walls 2x6), gypsum both sides; sound insulation at bedrooms, baths, laundry.
F1 FLOOR: concrete slab on grade over vapor retarder and capillary break (CALGreen 4.505.2); foundation, slab and footings per structural; floor level per civil.

Section cuts: A at the lounge / kitchen, B at the guest room, C along the lobby (see A-2). Heights above finished floor; existing grade approximate (survey, assumed floor 1435.1').

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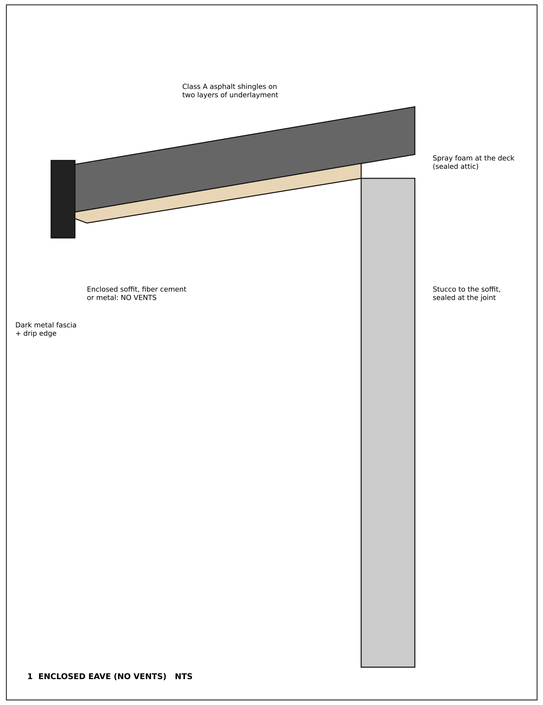
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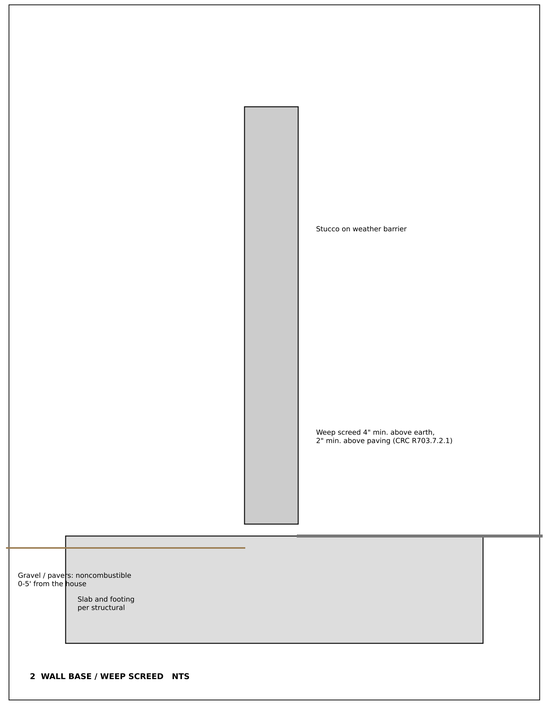
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BUILDING SECTIONS

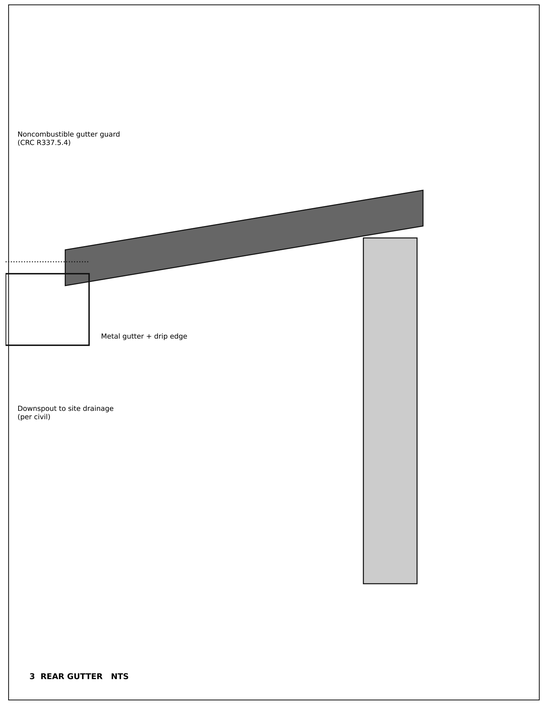
A-5



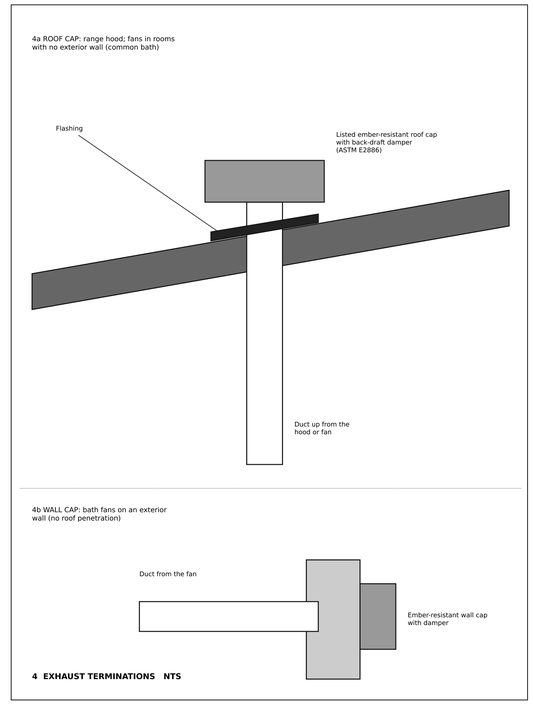
1 ENCLOSED EAVE (NO VENTS) NTS



2 WALL BASE / WEEP SCREED NTS



3 REAR GUTTER NTS



4 EXHAUST TERMINATIONS NTS

WILDLAND-URBAN INTERFACE REQUIREMENTS (VERY HIGH FIRE HAZARD SEVERITY ZONE)

CRC	ITEM	THIS PROJECT
R337.5	Roof	Class A asphalt shingles; eave and rake edges, ridge cap and the shed / gable junction closed and sealed so embers can't get under them. Carport: standing seam, Class A.
R337.5.4	Gutters	Noncombustible gutter guards on all gutters.
R337.6	Vents	No attic or eave vents (sealed attic). Exhaust and hood terminations: listed ember-resistant vents (ASTM E2886) or noncombustible 1/16"–1/8" mesh with dampers.
R337.7	Exterior walls	3-coat stucco (noncombustible). Porch cladding: noncombustible or ignition-resistant wood-look (fiber cement / porcelain).
R337.7.5	Eaves, soffits	Enclosed soffits, noncombustible or ignition-resistant (fiber cement or metal); fascia dark metal. Porch ceiling the same.
R337.8.2	Glazing	Dual-pane, at least one tempered pane, all windows, doors, skylight, clerestory.
R337.8.3	Exterior doors	Noncombustible, ignition-resistant, or solid core at least 1-3/8" thick.
R337.9	Decks	None. At-grade concrete landing and entry court only.
R337.10	Carport, fences	Carport roof Class A; framing within 5' of the house noncombustible or ignition-resistant. Fences and gates within 5' of the house noncombustible.
--	Zone 0 (0-5')	Noncombustible ground cover (gravel, pavers, concrete) within 5' of the house; no bark mulch or plants against the walls.

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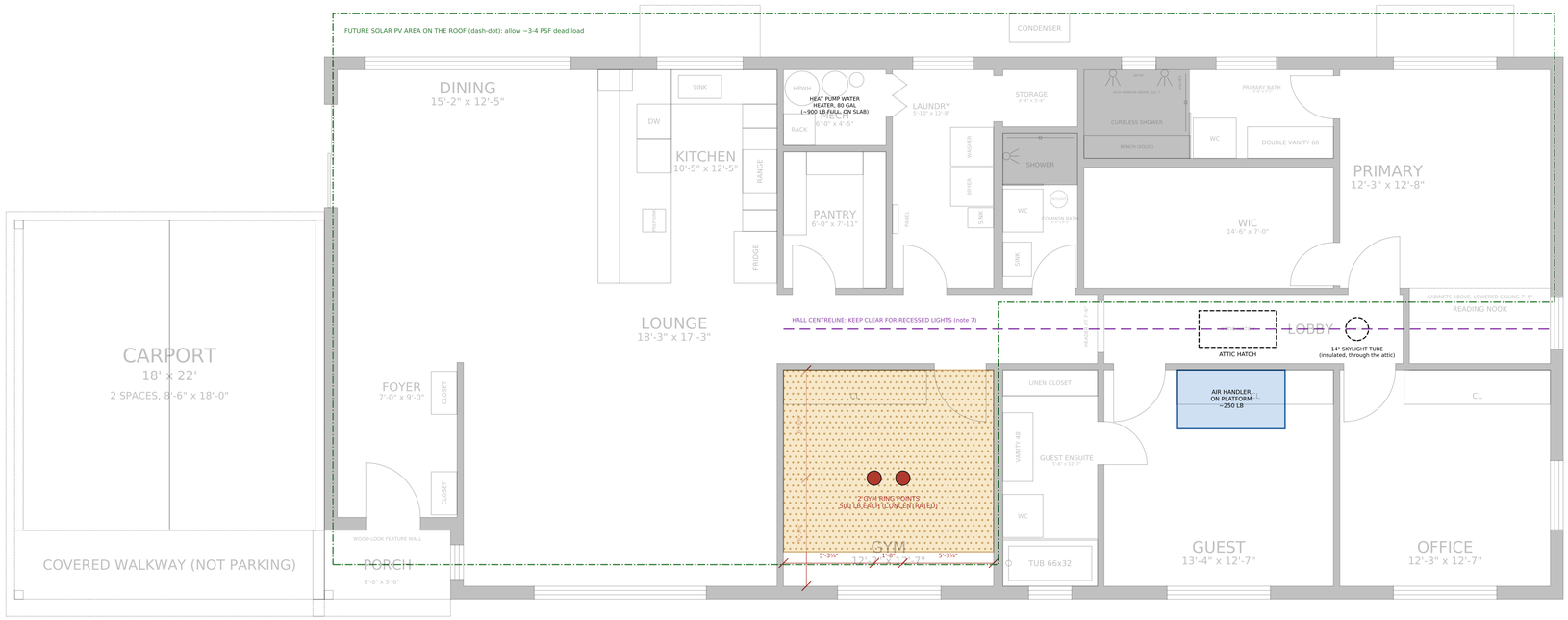
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**FIRE-ZONE
DETAILS
(CRC R337)**

A-6



ATTIC STORAGE (hatched): 20 PSF LIVE LOAD, PLYWOOD DECKED
(over the gym, under the shed: 2'-0" off the street wall to the hall wall)

ATTIC AND LOADS PLAN 1/4" = 1'-0"

LOAD NOTES (for the structural engineer and truss designer)

1. Gym ring points: two eye bolts 20" apart at the gym's centre, rings swinging along the room's long (street-to-hall) direction. 500 lb concentrated at each, dynamic: blocking across several trusses or joists with framing connectors; through-bolted forged eye bolts with plate washers.
2. Attic storage: 20 psf live load on the decked area shown (hatched), over the gym, where the shed roof gives the attic its height; light storage only. The rest of the wing's attic, under the gable, is tallest along the ridge.
3. Air handler: on a platform beside the attic hatch, near the ridge, with a 30" walkway, light and outlet (mechanical code).
4. Future solar: on the roof plane, clear of the fire setbacks (A-3); include its dead load in the roof design.
5. Heat pump water heater: on the slab in the mechanical closet.
6. Vaulted great room: no attic and no loads above the ceiling.
7. Bedroom hall ceiling: keep the hall's centreline free of blocking, strongbacks, ducts and pipes, so recessed lights fit between the trusses along it (clear of the hatch, smoke alarm and return grille). Run the duct trunk to one side. The 14" skylight takes the last spot, near the end; its tube jogs toward the street in the attic (two elbows) to its dome.

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Very High Fire Hazard Severity Zone

Owner: Richard Acosta

Prepared by: _____

Structural: by engineer (separate sheets)

DRAFT FOR REVIEW not for construction

Date: October 15, 2026
Revisions:

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ATTIC AND STRUCTURAL LOADS PLAN

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